# Kedar Joshi

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## EDUCATION

**University of Southern California, Los Angeles, CA** **Aug 2026 – May 2028**
MS in Computer Science

**Manipal University Jaipur, Jaipur, India** **Graduated May 2026**
Bachelor of Technology in Computer Science with *Distinction* (CGPA 8.88/10)
Relevant Courses: Data Structure and Algorithms, Artificial Intelligence and Soft Computing, Data Science and Machine Learning

## PUBLICATIONS

**The Misallocation of Precision in Quantized Language Models** **Preprint, 2026**
**Kedar Joshi**. *Sole author.*

- The activation-covariance Hessian trace used by GPTQ, OmniQuant and ResQ to allocate bit-width is anti-correlated with functional importance across all 64 sub-layers of Llama-3.1-8B ($\rho = -0.506$, $p < 0.001$): the layers those methods protect are the ones that matter least.
- Crossover dissociation under Benjamini-Hochberg FDR correction across 192 tests: no attention sub-layer reaches significance for ARC-Challenge, no MLP sub-layer for GSM8K. Replicates on Mistral-7B and DeepSeek-R1-Distill, and under GPTQ.
- A 6-of-32-layer mixed-precision model is statistically indistinguishable from FP16 on all three benchmarks at 42% of its memory; at an identical budget, Hessian-selected layers recover 2.6% of quantization damage against 52.6% for probe-selected layers ($p = 0.001$).

**Contrastive Activation Decomposition: Activation Steering Mitigation by Leveraging Bias Geometry** **Preprint, 2026**
**Kedar Joshi**, Kripabandhu Ghosh. *First author.*

- Diagnostic that prescribes intervention rank and target layers before any steered generation, by decomposing contrastive activation differences.
- Bias geometry varies by type: housing discrimination is near rank-1 ($EVR_1 = 67.9\%$), medicalization bias is diffuse ($EVR_1 = 10.2\%$), explaining why rank-1 steering methods fail on one and succeed on the other.
- Matches CAA's debiasing on AccessEval ($d = 0.337$ vs $0.341$) with no per-task tuning, at a fifth of the output-quality cost; validated on Llama, Qwen, Gemma, and Mistral models.

**Don't Judge a Book by its Cover: Testing LLMs' Robustness Under Logical Obfuscation** **EACL 2026 (main conference)**
Abhilekh Borah, Shubhra Ghosh*, **Kedar Joshi***, Aditya Kumar Guru*, Kripabandhu Ghosh. **Equal contribution.*

- Introduced Logifus, a structure-preserving logical obfuscation algorithm, and LogiQAte, a 1,108-question diagnostic benchmark across four reasoning tasks.
- Frontier models degrade by 28.8% on average under logically equivalent rewrites, up to −86% in the worst case, with a 50–80% collapse in late-layer next-token confidence.
- Code: github.com/abhilekhborah/logical-obfuscation · Data: huggingface.co/datasets/abhilekhborah/LogiQAte

**Integrating GLCM Texture Analysis for Improved Deepfake Detection on CelebDF(v2) Dataset** **ACOIT 2024, IEEE**
**Kedar Joshi**, Aditya Sinha. *First author.*

- Texture and frequency features with gradient-boosted trees, detecting manipulation without a CNN forward pass.
- Targets resource-constrained and edge deployment where CNN-based detectors are impractical.
- ieeexplore.ieee.org/document/10939531

## PROFESSIONAL EXPERIENCE

**Indian Institute of Science Education and Research (IISER), Kolkata, India** **May 2025 – Nov 2025**
*Research Intern – LLM Evaluation and Robustness*

- Evaluated six state-of-the-art models (GPT-4o, GPT-5, o4-mini, Claude, Gemini, Qwen) for reasoning robustness under logical obfuscation; published at EACL 2026 (see Publications).

**India Meteorological Department, Bhopal, India** **Nov 2024 – Feb 2025**
*Research Intern – AI/ML for Atmospheric Sciences*

- Filed a patent application for automated physics-informed neural architecture search for single-radar 3D atmospheric velocity field reconstruction.
- 3D wind retrieval has required two or more Doppler radars since the 1960s, because a single radar measures only the radial velocity component. This is the first method to reconstruct the full field from one radar, using conservation laws in place of the geometric constraints the second radar provided.
- Architecture search combining DARTS and Bayesian optimization, treating physics compliance as a third objective alongside accuracy and efficiency, across 245K to 50M+ parameters.

## ACADEMIC PROJECTS AND PAPERS

**Perturbation-Aware Fine-Tuning (PAFT) for Reducing LLM Memorization**

- Training-time data augmentation method that rewrites training examples to reduce memorization, with MemGenTOI, a metric for the memorization-generalization trade-off.

## TECHNICAL SKILLS

- **Languages & Tools:** Python, C, PyTorch, TensorFlow, CUDA, Git, Prover9
- **Areas:** LLM evaluation, mechanistic interpretability, quantization, physics-informed neural networks, neural architecture search, benchmark development